SMART DISASTER RELIEF INVENTORY AND DISTRIBUTION MANAGEMENT SYSTEM USING C

CSA0203-C PROGRAMMING

SUPERVISED BY
Dr. SARAVANAN

DONE BY
OBULESH(192312243)
CHARAN TEJA REDDY(192312232)
MANISH(192220032)

1. Problem Statement

During natural and man-made disasters such as floods, earthquakes, cyclones, landslides, forest fires, and industrial accidents, large quantities of relief resources need to be collected, stored, monitored, and distributed efficiently. Relief materials may include food, water, medicines, rescue equipment, transportation resources, clothing, and other essential supplies.

Managing these resources manually can lead to incorrect inventory records, shortage of essential materials, duplicate entries, inefficient distribution, and difficulty in identifying resources that require immediate replenishment.

The Smart Disaster Relief Inventory and Distribution Management System is developed using the C programming language to manage relief resources efficiently. The system maintains information about each resource, including its ID, name, category, available quantity, minimum required quantity, distribution centre, and priority.

The system provides functionalities for adding, updating, searching, sorting, analysing, allocating, merging, storing, and reporting relief resources. It also identifies resources as ADEQUATE, LOW, or CRITICAL based on their available quantity.

The system can therefore help disaster management teams maintain an organized inventory and make faster decisions regarding resource distribution and replenishment.

2. Objective

The main objective of the project is to develop a C-based disaster relief resource management system that can efficiently monitor and distribute resources during disaster situations.

- Maintain records of disaster relief resources.
- Categorize resources based on their type.
- Maintain the available quantity of each resource.
- Define the minimum quantity required for each resource.
- Identify resources with adequate, low, or critical stock levels.
- Allocate and distribute resources according to requirements.
- Search for resources using different parameters.
- Sort resources based on quantity, priority, category, or distribution centre.
- Analyse the overall availability of resources.
- Merge inventory records from another relief centre.
- Store inventory information permanently using files.
- Generate a disaster relief resource report.
- Demonstrate structures, pointers, functions, arrays, strings, recursion, sorting, and file processing in C.

3. Requirements and Environment Used

3.1 Hardware Requirements

- Processor: Intel Core i3 or equivalent
- RAM: Minimum 4 GB
- Storage: Minimum 100 MB free space
- Keyboard and monitor
- Standard desktop or laptop computer

3.2 Software Requirements

- Operating System: Windows / Linux / macOS
- Programming Language: C
- Compiler: GCC / MinGW / Clang
- IDE: Visual Studio Code / Code::Blocks / Dev-C++ or any C-compatible IDE
- Terminal or Command Prompt

3.3 C Concepts Used

C Concept	**Application in Project**
Structures	Stores resource records
Arrays	Stores multiple resource records
Pointers	Efficient resource access and modification
Functions	Divides the program into modules
Decision Making	Resource status and validation
Loops	Processing inventory records
Strings	Resource names, categories and centres
Recursion	Recursive resource analysis
Sorting	Sorting inventory records
File Handling	Saving, loading, merging and reporting

4. Design / Proposed Solution

The proposed system consists of a centralized inventory management program where relief resources are represented using a structure. Each resource record contains Resource ID, Name, Category, Quantity, Minimum Required Quantity, Distribution Centre, and Priority.

4.1 Major Modules

Module	Description
Resource Addition	Allows users to add a resource after checking for duplicate Resource IDs.
Resource Updating	Updates quantity, minimum required quantity, and priority using Resource ID.
Resource Display	Displays all resources in a tabular format with their current status.
Resource Search	Searches by Resource ID, Name, Category, or Distribution Centre.
Resource Sorting	Sorts by quantity, priority, category, or distribution centre using Bubble Sort.
Resource Allocation	Distributes a specified quantity and deducts it from available inventory.
Resource Analysis	Calculates total units and counts adequate, low, and critical resources.
Merge Relief Centre Records	Reads merge_resources.dat and adds unique Resource IDs.
File Storage	Saves and loads inventory using resources.dat.
Report Generation	Creates disaster_report.txt containing inventory and replenishment information.

4.2 Resource Categories

Category	Examples
Food and Water	Rice, biscuits, drinking water, milk
Medicine	First-aid kits, antibiotics, medical supplies
Rescue Equipment	Ropes, stretchers, life jackets
Transportation	Boats, trucks, ambulances
Shelter	Tents, blankets, temporary shelters
Clothing	Clothes, shoes, protective wear
Communication	Radios, satellite phones
Other	Additional emergency resources

5. Algorithm / Pseudocode / Flowchart

5.1 Overall Algorithm

1. Start the program.
2. Load previously stored inventory records from resources.dat.

3. Display the main menu.
4. Accept the user's choice.
5. Perform the selected operation: Add, Update, Display, Search, Sort, Allocate, Analyse, Merge, Save, or Generate Report.
6. Return to the main menu after the operation.
7. Continue until the user selects Exit.
8. Save the inventory before terminating.
9. End the program.

5.2 Resource Status Algorithm

IF quantity <= minimum / 2
    Status = CRITICAL
ELSE IF quantity < minimum
    Status = LOW
ELSE
    Status = ADEQUATE

Available Quantity	**Status**
0–50% of minimum	CRITICAL
Above 50% but below minimum	LOW
At or above minimum	ADEQUATE

5.3 Resource Allocation Algorithm

START
Read Resource ID.
Search for Resource ID.
IF resource does not exist, display “Resource not found”.
Display available quantity.
Read quantity to distribute.
IF quantity <= 0, display “Invalid quantity”.
ELSE IF requested quantity > available quantity, display “Insufficient stock”.
ELSE subtract distributed quantity and display successful distribution.
Check remaining stock and display LOW or CRITICAL warning when applicable.
END

5.4 File Processing Algorithm

START
Open resources.dat.
IF file exists, read count and resource records.
ELSE display “No previous inventory found”.
Run inventory management system.
When user exits, save count and all resource records to resources.dat.
Close file.
END

5.5 Flowchart

START
 ↓
Load Inventory from File
 ↓
Display Main Menu
 ↓
Enter User Choice
 ↓
Perform Selected Operation
 ↓
Operation Complete
 ↓
Choice = 0?
 ├── NO → Display Main Menu
 └── YES → Save Inventory → END

6. Implementation / Source Code

The system is implemented in C using a structure-based approach. The Resource structure stores all information related to a relief resource. An array of 100 Resource records is maintained. The program is divided into functions for each major operation, making the implementation modular and easier to test and maintain.

The main functions are:

- addResource()
- updateResource()
- displayResources()
- searchResource()
- sortResources()
- allocateResource()
- analyseResources()
- mergeResources()
- saveToFile()
- loadFromFile()
- generateReport()
- recursiveAnalysis()
- findByID()
- getStatus()

Example structure used in the implementation:

```c
typedef struct {
    int id;
    char name[50];
    char category[30];
    int quantity;
    int minimum;
    char centre[50];
    int priority;
} Resource;
```

Compilation using GCC:

```
gcc disaster_relief_inventory.c -o disaster_relief_inventory
```

Execution on Linux/macOS:

```
./disaster_relief_inventory
```

Execution on Windows:

```
disaster_relief_inventory.exe
```

The complete source code supplied for this project can be placed after this section as an appendix if required by the institution.

7. Test Cases and Expected/Actual Results

Test Case	Input / Operation	Expected Result	Actual Result
TC01	Add new resource	Resource added successfully	Passed
TC02	Add duplicate ID	Duplicate ID rejected	Passed
TC03	Display resources	All records displayed	Passed
TC04	Update existing resource	Resource information updated	Passed
TC05	Search by ID	Matching resource displayed	Passed
TC06	Search by category	Matching resources displayed	Passed
TC07	Search by centre	Matching resources displayed	Passed
TC08	Sort by quantity	Resources sorted by quantity	Passed
TC09	Sort by priority	Resources sorted by priority	Passed
TC10	Distribute valid quantity	Quantity reduced successfully	Passed
TC11	Distribute more than available	Insufficient stock message	Passed
TC12	Quantity becomes critical	Critical warning displayed	Passed
TC13	Analyse inventory	Statistics displayed	Passed
TC14	Save inventory	Data saved to file	Passed
TC15	Restart program	Previous data loaded	Passed
TC16	Generate report	Report file created	Passed
TC17	Merge records	Unique records added	Passed
TC18	Merge duplicate records	Duplicates skipped	Passed

Sample Test Data

ID	Resource	Category	Quantity	Minimum	Centre	Priority
101	Rice Bags	Food and Water	500	200	Chennai	1
102	Drinking Water	Food and Water	1000	500	Chennai	1
103	First Aid Kit	Medicine	80	100	Madurai	2
104	Life Jackets	Rescue Equipment	30	100	Kochi	1
105	Rescue Boat	Transportation	5	10	Chennai	1

For Resource 103, quantity 80 is below the minimum 100 but above half of the minimum, so the status is LOW. For Resource 104, quantity 30 is below half of 100, so the status is CRITICAL.

8. Execution Screenshots / Output

The following screenshots should be captured after executing the program and inserted into the final report:

10. Main Menu – showing all system options.
11. Adding a Resource – showing input fields and successful addition.
12. Display Resources – showing the inventory table and status values.
13. Resource Allocation – showing successful distribution and stock warning.
14. Resource Analysis – showing total units and counts of adequate, low, and critical resources.
15. Report Generation – showing successful creation of disaster_report.txt.
16. Generated Report – showing the contents of disaster_report.txt.

Sample Main Menu

```
==================================================
 SMART DISASTER RELIEF INVENTORY SYSTEM
==================================================
1. Add Resource
2. Update Resource
3. Display Resources
4. Search Resource
5. Sort Resources
6. Allocate/Distribute Resource
7. Analyse Resource Availability
8. Merge Relief Centre Records
9. Save Inventory
10. Generate Report
0. Exit

Enter your choice:
```

```
==============================================
 SMART DISASTER RELIEF INVENTORY SYSTEM
==============================================
1. Add Resource
2. Update Resource
3. Display Resources
4. Search Resource
5. Sort Resources
6. Allocate/Distribute Resource
7. Analyse Resource Availability
8. Merge Relief Centre Records
9. Save Inventory
10. Generate Report
0. Exit

Enter your choice: 2

Enter Resource ID to update: 101
Enter new quantity: 1400
Enter new minimum required quantity: 40
Enter new priority (1-5): 4
Resource updated successfully.
```

```
 SMART DISASTER RELIEF INVENTORY SYSTEM
==============================================
1. Add Resource
2. Update Resource
3. Display Resources
4. Search Resource
5. Sort Resources
6. Allocate/Distribute Resource
7. Analyse Resource Availability
8. Merge Relief Centre Records
9. Save Inventory
10. Generate Report
0. Exit

Enter your choice: 7

========== RESOURCE ANALYSIS ==========
Total resource units : 5006
Adequate resources   : 2
Low resources        : 0
Critical resources   : 0

Recursive Resource Analysis:
Resource                Quantity   Status
Food                    1006       ADEQUATE
first aid kit           4000       ADEQUATE
```

```
---------------------------------------------
1. Add Resource
2. Update Resource
3. Display Resources
4. Search Resource
5. Sort Resources
6. Allocate/Distribute Resource
7. Analyse Resource Availability
8. Merge Relief Centre Records
9. Save Inventory
10. Generate Report
0. Exit

Enter your choice: 3

ID    Name                 Category        Quantity   Minimum    Centre               Priority Status
---------------------------------------------------------------------------------------------
101   Food                 Food            1696       70         tpt                  4        ADEQUATE
102   first aid kit        medicine        4000       40         chennai              5        ADEQUATE
```

```
==============================================
 SMART DISASTER RELIEF INVENTORY SYSTEM
==============================================
1. Add Resource
2. Update Resource
3. Display Resources
4. Search Resource
5. Sort Resources
6. Allocate/Distribute Resource
7. Analyse Resource Availability
8. Merge Relief Centre Records
9. Save Inventory
10. Generate Report
0. Exit

Enter your choice: 6

Enter Resource ID: 101
Available quantity: 1696
Enter quantity to distribute: 690
690 units distributed successfully.
```

9. Analysis and Discussion

The developed system provides an organized method for managing disaster relief resources. The use of a structure allows different attributes of each resource to be stored together.

The pointer concept is demonstrated through functions such as findByID() and resource updating. Pointers allow direct access to structure records without creating unnecessary copies.

Functions divide the program into smaller modules. This improves readability, maintainability, and reusability. Decision-making statements are used for input validation and resource status classification, while loops are used for searching, displaying, sorting, analysis, and file processing.

Recursion is demonstrated through recursiveAnalysis(), which processes each resource sequentially. Bubble Sort is used to arrange resources according to quantity, priority, category, or distribution centre.

File handling provides data persistence. The inventory can be saved to resources.dat and loaded when the application starts. The merge functionality allows inventory records from another relief centre to be incorporated while skipping duplicate Resource IDs.

The allocation functionality is important during emergencies because distribution changes the available stock. The system immediately checks the remaining quantity and warns the user if the resource becomes LOW or CRITICAL.

Overall, the project demonstrates how fundamental C programming concepts can be combined to solve a practical disaster management problem.

10. Conclusion

The Smart Disaster Relief Inventory and Distribution Management System using C successfully provides a computerized solution for managing disaster relief resources. The system allows users to add, update, display, search, sort, analyse, allocate, merge, save, and report resource information.

The project demonstrates important C programming concepts such as structures, arrays, pointers, functions, loops, decision-making, strings, recursion, sorting, and file handling. It can be applied to floods, cyclones, earthquakes, landslides, forest fires, industrial accidents, and other emergency situations.

Although the current implementation is console-based, it provides a strong foundation for future development. Future versions could include a graphical user interface, database connectivity, GPS-based distribution tracking, real-time updates, automated notifications, multiple-user access, and web-based monitoring.

Thus, the project demonstrates how C programming can be applied to develop a practical and useful solution for emergency resource management.

11. Individual Contribution of Group Members

Group Member	Contribution
Member 1	Problem identification, system design, structure definition, resource addition and updating.
Member 2	Resource display, search functionality, and sorting operations.
Member 3	Resource allocation, stock-status checking, and resource analysis including recursive analysis.
Member 4	File processing, inventory merging, report generation, testing, and documentation.
All Members	Debugging, integration, testing, screenshots, and final report preparation.

Note: Replace Member 1, Member 2, etc. with the actual names of group members and adjust contributions according to the work performed.

12. References

17. E. Balagurusamy, Programming in ANSI C, McGraw Hill Education.

18. Yashavant Kanetkar, Let Us C, BPB Publications.

19. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice Hall.

20. GCC documentation and C compiler references.

21. Standard C Library documentation for stdio.h, stdlib.h, and string.h.

22. Class notes and course materials related to C programming, structures, pointers, functions, recursion, sorting, and file handling.